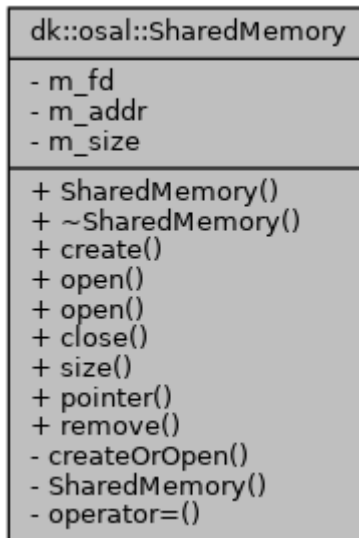


dk::osal::SharedMemory Class Reference

```
#include <DKOsa1SharedMemory.h>
```

Collaboration diagram for dk::osal::SharedMemory:



Public Member Functions

SharedMemory ()

Default Constructor. [More...](#)

~SharedMemory ()

Default Destructor. [More...](#)

Error **create** (const char *name, uint32_t **size**)

Create or Open an existing shared memory block. [More...](#)

Error **open** (const char *name)

Error **open** (const char *name, uint32_t **size**)

Open an existing shared memory block with a defined size.

If the shm block could be opened it will be truncated, because the creating side might not yet be finished with its **ftruncate()**. So we do it on both sides to be sure, the shm block has the correct size.

See also <http://stackoverflow.com/q/16502767/3519749>. [More...](#)

void **close** ()

Closes the mapping of shared memory block. [More...](#)

uint32_t **size** () const

Get the size of the currently mapped shared memory block. [More...](#)

void * **pointer** () const

Get a pointer to the shared memory block. [More...](#)

Static Public Member Functions

static **Error** **remove** (const char *name)
Remove (unlink) the name of the shared memory block. [More...](#)

Private Member Functions

Error **createOrOpen** (const char *name, uint32_t **size**, bool doCreate)
Helper method during the real work. Creates or Opens a shared memory region depending on doCreate. [More...](#)

SharedMemory (const **SharedMemory** &)
Disable Copy Constructor. [More...](#)

SharedMemory & **operator=** (const **SharedMemory** &)
Disable assignment operator. [More...](#)

Private Attributes

int64_t **m_fd**
char * **m_addr**
uint32_t **m_size**

Constructor & Destructor Documentation

◆ SharedMemory() [1 / 2]

dk::osal::SharedMemory::SharedMemory ()

Default Constructor.

requirement ID 751539

◆ ~SharedMemory()

dk::osal::SharedMemory::~~SharedMemory ()

Default Destructor.

requirement ID 751539

◆ SharedMemory() [2/2]

```
dk::osal::SharedMemory::SharedMemory ( const SharedMemory & )
```

private

Disable Copy Constructor.

Member Function Documentation

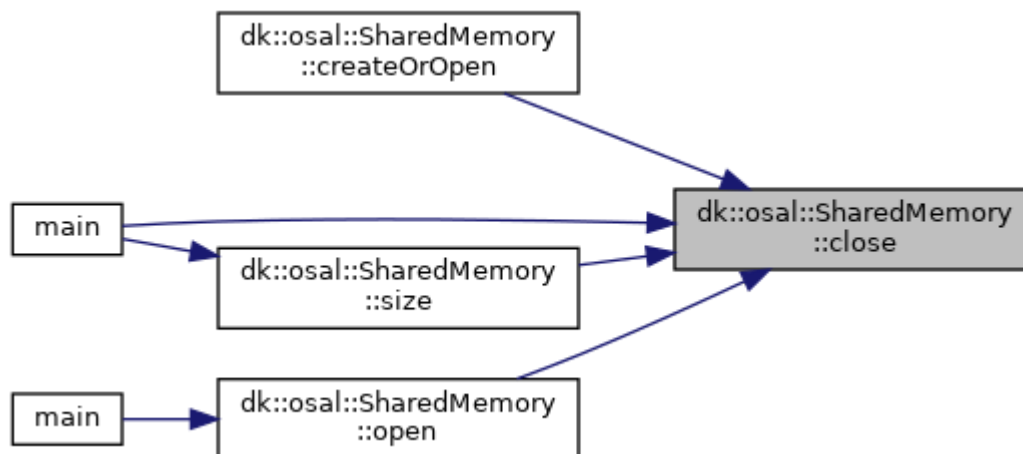
◆ close()

```
void dk::osal::SharedMemory::close ( )
```

Closes the mapping of shared memory block.

requirement ID 751595

Here is the caller graph for this function:



◆ create()

```
dk::osal::SharedMemory::create ( const char * name,  
                                uint32_t size  
                                )
```

Create or Open an existing shared memory block.

Parameters

name Global identifier for shm block. name can be upto NAME_MAX (255) characters long. Under QNX:
The name is interpreted as follows:

"entry"	"\$CWD/entry"
"/entry"	"/dev/shmem/entry"
"entry/entry2"	"\$CWD/entry/entry2"
"/entry/entry2"	"/entry/entry2"

For portable use, you should use the form "/entry".

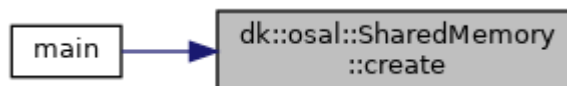
size size of the shm block.

Returns

Error code.

tequirement id 751593

Here is the caller graph for this function:



◆ createOrOpen()

```

Error dk::osal::SharedMemory::createOrOpen ( const char * name,
                                              uint32_t    size,
                                              bool        doCreate
                                              )

```

private

Helper method during the real work. Creates or Opens a shared memory region depending on doCreate.

Parameters

- name** global identifier for shm block.
- size** size of the shm block. For doCreate=false this can be 0. however you might run into race condition then (see description of [open\(\)](#))
- doCreate** true for creating the shm object, false for just opening it.

Returns

Error Code.

requirement id 751593,751594

Here is the call graph for this function:



◆ open() [1/2]

```

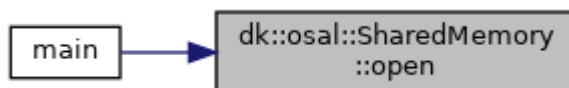
Error dk::osal::SharedMemory::open ( const char * name )

```

Here is the call graph for this function:



Here is the caller graph for this function:



◆ open() [2/2]

```
dk::osal::SharedMemory::open ( const char * name,
                                uint32_t    size
                                )
```

Open an existing shared memory block with a defined size.

If the shm block could be opened it will be truncated, because the creating side might not yet be finished with its **ftruncate()**. So we do it on both sides to be sure, the shm block has the correct size. See also <http://stackoverflow.com/q/16502767/3519749>.

Parameters

name Global identifier for shm block.

size Size of shm block.

Returns

Error code.

requirement ID 751594

◆ operator=()

SharedMemory & dk::osal::SharedMemory::operator= (const **SharedMemory** &)

private

Disable assignment operator.

◆ pointer()

```
void * dk::osal::SharedMemory::pointer ( ) const
```

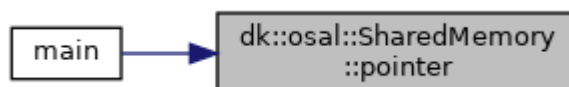
Get a pointer to the shared memory block.

Returns

the pointer

requirement id 751597

Here is the caller graph for this function:



◆ remove()

```
static Error dk::osal::SharedMemory::remove ( const char * name )
```

static

Remove (unlink) the name of the shared memory block.

Parameters

name Global identifier for shm block.

Returns

Error code.

requirement id 751598

◆ size()

```
uint32_t dk::osal::SharedMemory::size ( ) const
```

Get the size of the currently mapped shared memory block.

Returns

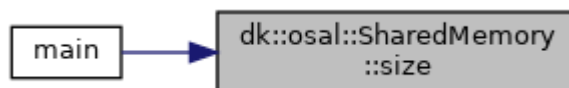
size of the shm block.

requirement id 751596

Here is the call graph for this function:



Here is the caller graph for this function:



Member Data Documentation

◆ m_addr

```
char* dk::osal::SharedMemory::m_addr
```

private

◆ m_fd

int64_t dk::osal::SharedMemory::m_fd

private

◆ m_size

uint32_t dk::osal::SharedMemory::m_size

private

The documentation for this class was generated from the following files:

- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsaiSharedMemory.h**
- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsaiSharedMemory.cpp**